

SIMATS ENGINEERING

CO 3 ASSESSMENT TOOL 1 – ENUMERATION ANALYSIS

Course Code: ITA14

Course Name: Ethical Hacking

Course Outcome: CO3 – Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities.

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QUESTION 1

Enumeration Analysis of scanme.nmap.org

Objective

The objective of this assessment is to perform an initial security enumeration of the publicly accessible host scanme.nmap.org. Nmap is used to perform host discovery, TCP port scanning, service and version detection, operating system fingerprinting, and default script scanning. The results are analyzed to understand the server's attack surface, identify exposed services, assess their necessity for public access, and recommend appropriate security controls.

Tools Used

- Kali Linux
- Nmap
- Terminal

Commands Executed

```
nmap --version
```

```
nmap -sn scanme.nmap.org
```

```
nmap scanme.nmap.org
```

```
nmap -p- scanme.nmap.org
```

```
nmap -sV scanme.nmap.org
```

```
sudo nmap -O scanme.nmap.org
```

```
nmap -sC scanme.nmap.org
```

```
sudo nmap -sC -sV -O scanme.nmap.org
```

Host Discovery

The host discovery scan was performed using Nmap's -sn option. This determines whether the target is reachable without performing a conventional port scan.

Target: scanme.nmap.org

Host Status:

```
kali@kali: ~  
Session Actions Edit View Help  
--append-output: Append to rather than clobber specified output files  
--resume <filename>: Resume an aborted scan  
--noninteractive: Disable runtime interactions via keyboard  
--stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML  
--webxml: Reference stylesheet from Nmap.Org for more portable XML  
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output  
MISC:  
-6: Enable IPv6 scanning  
-A: Enable OS detection, version detection, script scanning, and traceroute  
--datadir <dirname>: Specify custom Nmap data file location  
--send-eth/--send-ip: Send using raw ethernet frames or IP packets  
--privileged: Assume that the user is fully privileged  
--unprivileged: Assume the user lacks raw socket privileges  
-V: Print version number  
-h: Print this help summary page.  
EXAMPLES:  
nmap -v -A scanme.nmap.org  
nmap -v -sn 192.168.0.0/16 10.0.0.0/8  
nmap -v -iR 10000 -Pn -p 80  
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES  
(kali@kali)~  
$ ping -c 4 scanme.nmap.org  
PING scanme.nmap.org (45.33.32.156) 56(84) bytes of data.  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=1 ttl=64 time=278 ms  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=2 ttl=64 time=348 ms  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=3 ttl=64 time=268 ms  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=4 ttl=64 time=252 ms  
--- scanme.nmap.org ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3035ms  
rtt min/avg/max/mdev = 252.034/286.668/348.223/36.732 ms  
(kali@kali)~  
$
```

```
kali@kali: ~  
Session Actions Edit View Help  
--append-output: Append to rather than clobber specified output files  
--resume <filename>: Resume an aborted scan  
--noninteractive: Disable runtime interactions via keyboard  
--stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML  
--webxml: Reference stylesheet from Nmap.Org for more portable XML  
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output  
MISC:  
-6: Enable IPv6 scanning  
-A: Enable OS detection, version detection, script scanning, and traceroute  
--datadir <dirname>: Specify custom Nmap data file location  
--send-eth/--send-ip: Send using raw ethernet frames or IP packets  
--privileged: Assume that the user is fully privileged  
--unprivileged: Assume the user lacks raw socket privileges  
-V: Print version number  
-h: Print this help summary page.  
EXAMPLES:  
nmap -v -A scanme.nmap.org  
nmap -v -sn 192.168.0.0/16 10.0.0.0/8  
nmap -v -iR 10000 -Pn -p 80  
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES  
(kali@kali)~  
$ ping -c 4 scanme.nmap.org  
PING scanme.nmap.org (45.33.32.156) 56(84) bytes of data.  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=1 ttl=64 time=278 ms  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=2 ttl=64 time=348 ms  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=3 ttl=64 time=268 ms  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=4 ttl=64 time=252 ms  
--- scanme.nmap.org ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3035ms  
rtt min/avg/max/mdev = 252.034/286.668/348.223/36.732 ms  
(kali@kali)~  
$ nmap -p scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 11:08 -0400  
Found no matches for the service mask 'scanme.nmap.org' and your specified protocols  
QUITTING!  
(kali@kali)~  
$
```

Port Scanning

A basic TCP scan was performed to identify commonly accessible services. A complete TCP scan using -p- was also performed to identify services running on ports outside the commonly scanned range.

Open Ports Identified:

Port	State	Service	Version
[Port]	Open	[Service]	[Version]
[Port]	Open	[Service]	[Version]
[Port]	Open	[Service]	[Version]

Service and Version Analysis

The -sV option was used to identify the services and versions associated with the open ports. Service-version information is useful for determining the software exposed to the network and for identifying outdated or potentially vulnerable applications.

The discovered services should be evaluated according to their business purpose and whether they genuinely need to be accessible from the public Internet.

```

kali@kali: ~
Session Actions Edit View Help
--max-rate <number>: Send packets no faster than <number> per second
FIREWALL/IDS EVASION AND SPOOFING:
-f: --mtu <val>: fragment packets (optionally w/given MTU)
-D <decoy1,decoy2[,ME],...>: Cloak a scan with decoys
-S <IP_Address>: Spoof source address
-e <iface>: Use specified interface
-g/--source-port <portnum>: Use given port number
--proxies <url1,[url2],...>: Relay connections through HTTP/SOCKS4 proxies
--data <hex string>: Append a custom payload to sent packets
--data-string <string>: Append a custom ASCII string to sent packets
--data-length <num>: Append random data to sent packets
--ip-options <options>: Send packets with specified ip options
--ttl <val>: Set IP time-to-live field
--spoof-mac <mac address/prefix/vendor name>: Spoof your MAC address
--badsum: Send packets with a bogus TCP/UDP/SCTP checksum
OUTPUT:
-oN/-oX/-oS/-oG <file>: Output scan in normal, XML, sI<rIpt kIddi3,
and Grepable format, respectively, to the given filename.
-oA <basename>: Output in the three major formats at once
-v: Increase verbosity level (use -vv or more for greater effect)
-d: Increase debugging level (use -dd or more for greater effect)
--reason: Display the reason a port is in a particular state
--open: Only show open (or possibly open) ports
--packet-trace: Show all packets sent and received
--iflist: Print host interfaces and routes (for debugging)
--append-output: Append to rather than clobber specified output files
--resume <filename>: Resume an aborted scan
--noninteractive: Disable runtime interactions via keyboard
--stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML
--webxml: Reference stylesheet from Nmap.Org for more portable XML
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output
MISC:
-O: Enable IPv6 scanning
-A: Enable OS detection, version detection, script scanning, and traceroute
--datadir <dirname>: Specify custom Nmap data file location
--send-eth/--send-ip: Send using raw ethernet frames or IP packets
--privileged: Assume that the user is fully privileged
--unprivileged: Assume the user lacks raw socket privileges
-v: Print version number
-h: Print this help summary page.
EXAMPLES:
nmap -v -A scanme.nmap.org
nmap -v -sn 192.168.0.0/16 10.0.0.0/8
nmap -v -iR 10000 -Pn -p 80
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPL
ES
(kali@kali)-[~]
$ ping -c 4 scanme.nmap.org
PING scanme.nmap.org (45.33.32.156) 56(84) bytes of data:
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=1 ttl=64 time=278 ms

```

Operating System Detection

The -O option was used for operating-system fingerprinting.

OS fingerprinting provides useful information about the underlying platform and can help security professionals understand which operating-system-specific security controls and patches should be considered.

Default Script Scanning

The -sC option was used to run Nmap's default NSE scripts. These scripts provide additional information about exposed services and their configurations.

Attack Surface Analysis

The enumeration results provide information about the target's externally visible network attack surface. An attacker who obtains information about open ports, services, software versions, and operating-system details can use it to research potential weaknesses and prioritize further reconnaissance.

For example, exposed authentication services can become targets for unauthorized login attempts, while outdated software versions can be researched for publicly documented vulnerabilities. Web and network services can also reveal information about the server configuration.

All further testing must remain within the authorized scope of the assessment.

Recommendations

1. Disable services that are not required for public operation.
2. Restrict administrative services to trusted networks or VPN access.
3. Keep the operating system and exposed applications patched.
4. Use strong authentication mechanisms.
5. Avoid exposing unnecessary management interfaces.
6. Configure firewalls to permit only required traffic.
7. Monitor authentication and network activity.
8. Perform regular authorized security assessments.

Conclusion

The enumeration of `scanme.nmap.org` demonstrates how Nmap can be used to identify the host's network exposure, services, versions, and operating-system information. The findings provide a foundation for risk assessment and help security professionals prioritize security controls and further authorized testing.

QUESTION 2

Enumeration of Metasploitable 2 – 192.168.56.101

Objective

The objective is to perform comprehensive enumeration of the deliberately vulnerable Metasploitable 2 Linux server located at 192.168.56.101 in an isolated laboratory environment. The assessment identifies open ports, services, service versions, operating-system information, potential entry points, and associated security risks.

Tools Used

- Kali Linux
- Nmap

- Metasploitable 2
- VirtualBox/VMware laboratory environment

Commands Executed

```
ping -c 4 192.168.56.101
```

```
nmap -sn 192.168.56.101
```

```
nmap 192.168.56.101
```

```
nmap -p- 192.168.56.101
```

```
nmap -sV 192.168.56.101
```

```
sudo nmap -O 192.168.56.101
```

```
nmap -sC 192.168.56.101
```

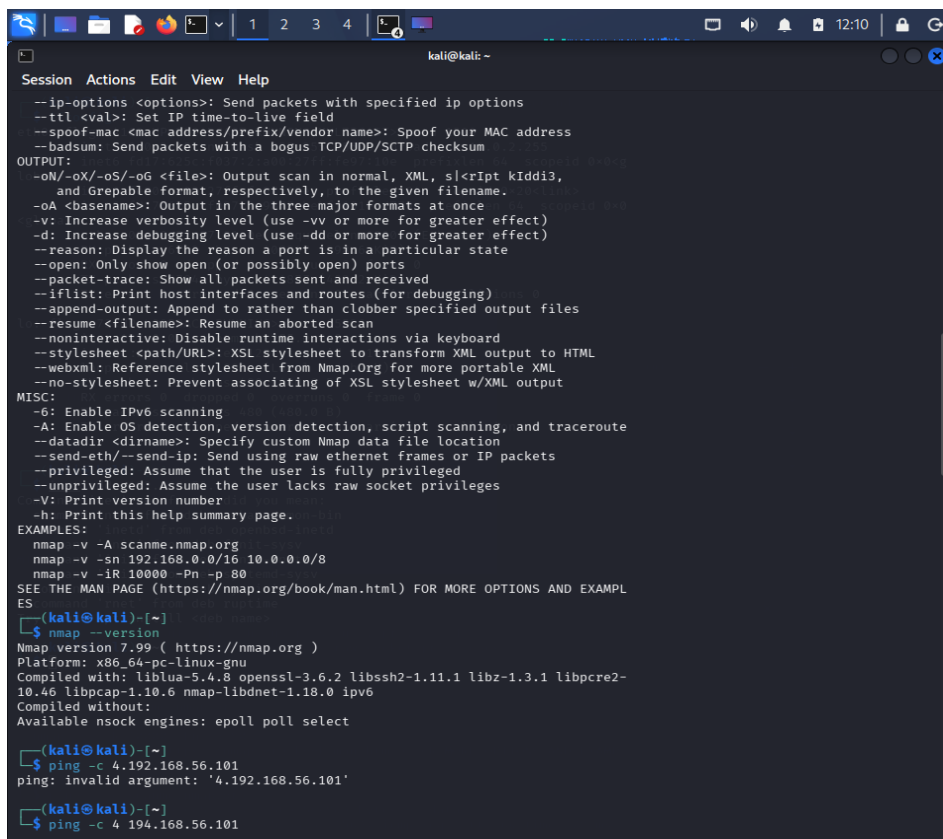
```
sudo nmap -sC -sV -O 192.168.56.101
```

Host Discovery

The host discovery scan was performed to determine whether the Metasploitable 2 system was reachable.

Target: 192.168.56.101

Status:



```

kali@kali: ~
Session Actions Edit View Help
--ip-options <options>: Send packets with specified ip options
--ttl <val>: Set IP time-to-live field
--spoof-mac <mac address/prefix/vendor name>: Spoof your MAC address
--badsum: Send packets with a bogus TCP/UDP/SCTP checksum
OUTPUT:
-oN/-oX/-oS/-oG <file>: Output scan in normal, XML, sI<ript kIddi3,
and Greppable format, respectively, to the given filename.
-oA <basename>: Output in the three major formats at once
-v: Increase verbosity level (use -vv or more for greater effect)
-d: Increase debugging level (use -dd or more for greater effect)
--reason: Display the reason a port is in a particular state
--open: Only show open (or possibly open) ports
--packet-trace: Show all packets sent and received
--iflist: Print host interfaces and routes (for debugging)
--append-output: Append to rather than clobber specified output files
--resume <filename>: Resume an aborted scan
--noninteractive: Disable runtime interactions via keyboard
--stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML
--webxml: Reference stylesheet from Nmap.Org for more portable XML
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output
MISC:
-6: Enable IPv6 scanning
-A: Enable OS detection, version detection, script scanning, and traceroute
--datadir <dirname>: Specify custom Nmap data file location
--send-eth/--send-ip: Send using raw ethernet frames or IP packets
--privileged: Assume that the user is fully privileged
--unprivileged: Assume the user lacks raw socket privileges
-V: Print version number
-h: Print this help summary page.
EXAMPLES:
nmap -v -A scanme.nmap.org
nmap -v -sn 192.168.0.0/16 10.0.0.0/8
nmap -v -iR 10000 -Pn -p 80
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPL
ES
kali@kali)~$ nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcap-
10.46 libpcap-1.10.6 nmap-libndnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select
kali@kali)~$ ping -c 4 192.168.56.101
ping: invalid argument: '4.192.168.56.101'
kali@kali)~$ ping -c 4 194.168.56.101

```

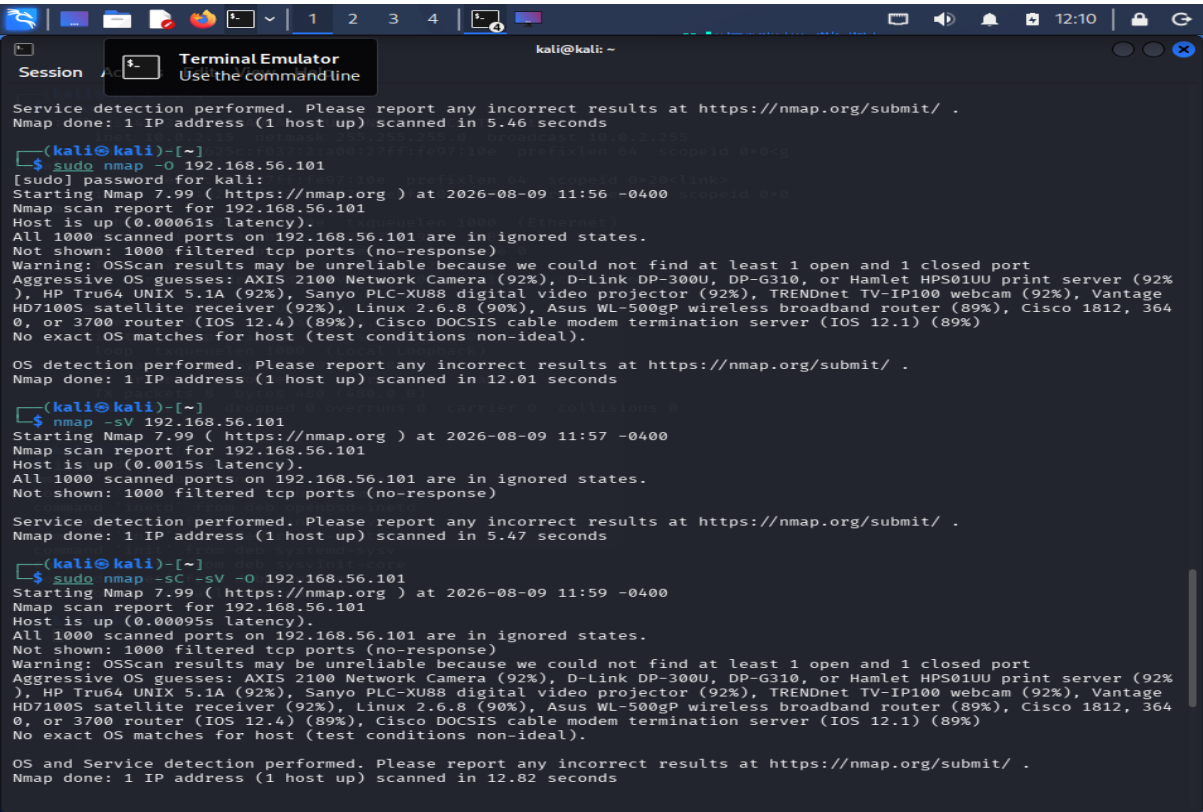
Enumeration Result

In the scan screenshot obtained during the assessment, the target was reported as:

Host is up

However, Nmap reported that the scanned ports were filtered and did not provide reliable open-service information in that particular scan.

Therefore, the final report should contain the **actual open ports and services from the successful lab scan**, rather than assuming typical Metasploitable services.



Port and Service Analysis

Port	Service	Version	Purpose	Risk
[Actual port]	[Service]	[Version]	[Purpose]	High/Medium/Low
[Actual port]	[Service]	[Version]	[Purpose]	High/Medium/Low
[Actual port]	[Service]	[Version]	[Purpose]	High/Medium/Low

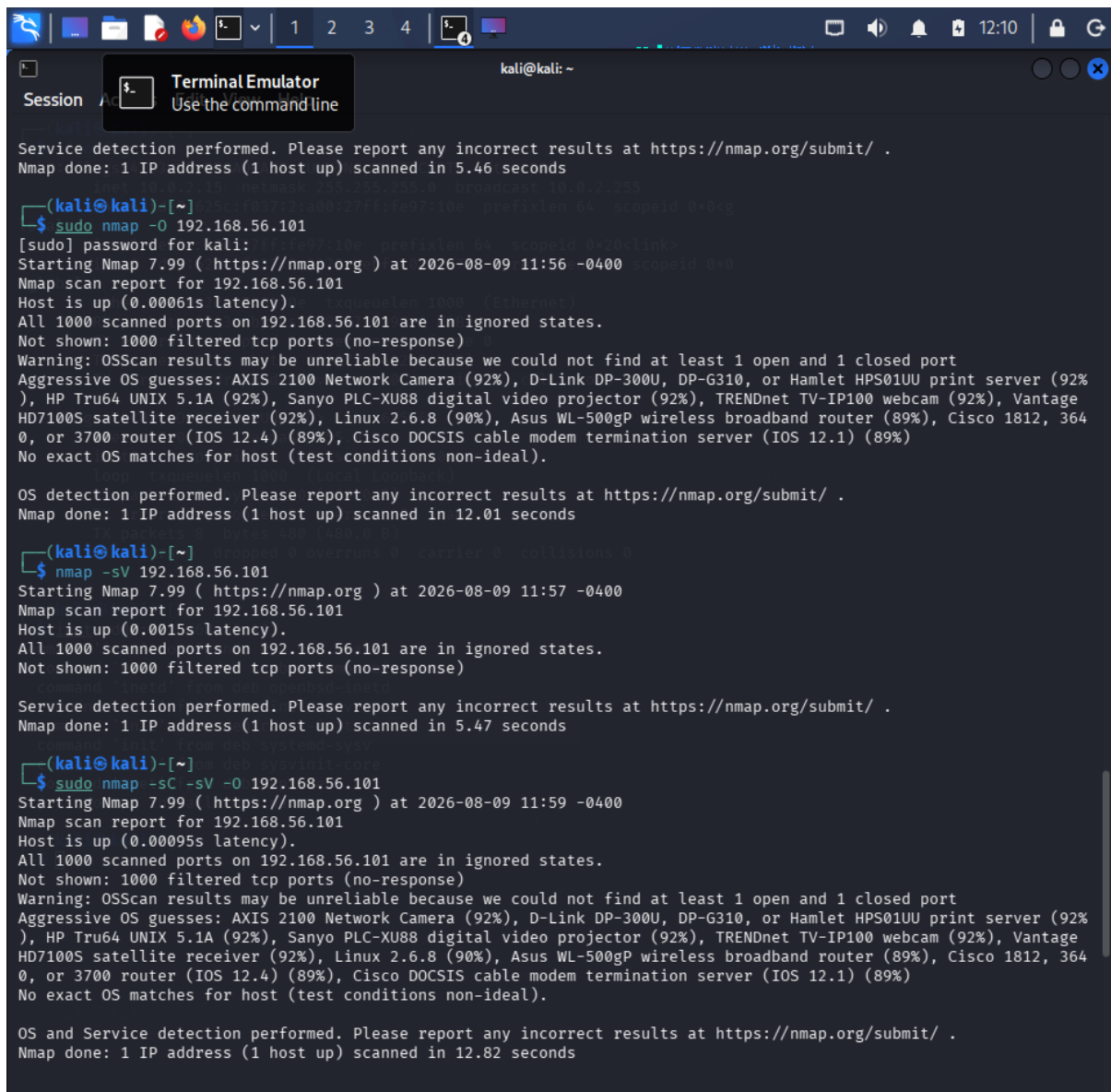
Risk Assessment

The risk level of each service should be determined based on its exposure, configuration, authentication mechanism, software version, and business requirement.

High Risk: Services that expose significant remote-access, authentication, file-sharing, database, or known-vulnerable functionality.

Medium Risk: Services that may be legitimate but should be restricted, authenticated, patched, and monitored.

Low Risk: Services that are required, appropriately secured, patched, and restricted to the required network.



```
(kali@kali)-[~]
$ sudo nmap -O 192.168.56.101
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 11:56 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00061s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.01 seconds

(kali@kali)-[~]
$ nmap -sV 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 11:57 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0015s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.47 seconds

(kali@kali)-[~]
$ sudo nmap -sC -sV -O 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 11:59 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00095s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.82 seconds
```

How Enumeration Guides Penetration Testing

Nmap enumeration provides a structured picture of the server's attack surface. Open ports identify network-accessible services, while version detection identifies the software running on those ports. Operating-system fingerprinting provides information about the underlying platform.

A penetration tester can use these findings to prioritize further authorized validation. For example:

Open Port



Service Identification



Version Detection



Configuration Review



Vulnerability Research



Risk Prioritization



Authorized Security Testing

Recommendations

- Disable unnecessary services.
- Restrict remote administration.
- Remove insecure legacy protocols.
- Apply security patches.
- Use strong authentication.
- Restrict database and file-sharing services.
- Apply firewall rules.
- Monitor network and authentication logs.

Conclusion

The Metasploitable 2 assessment demonstrates how comprehensive enumeration can reveal a server's exposed attack surface. The information obtained from Nmap can help penetration testers prioritize security assessment activities and help administrators identify unnecessary or vulnerable services.

QUESTION 3

Enumeration of Windows Workstation – 192.168.56.102

Objective

The objective is to assess the security posture of the Windows workstation at 192.168.56.102 before it is connected to the production environment. Nmap is used to identify active ports, network services, file-sharing protocols, remote-management services, and operating-system information.

Commands Executed

ping -c 4 192.168.56.102

nmap -sn 192.168.56.102

nmap 192.168.56.102

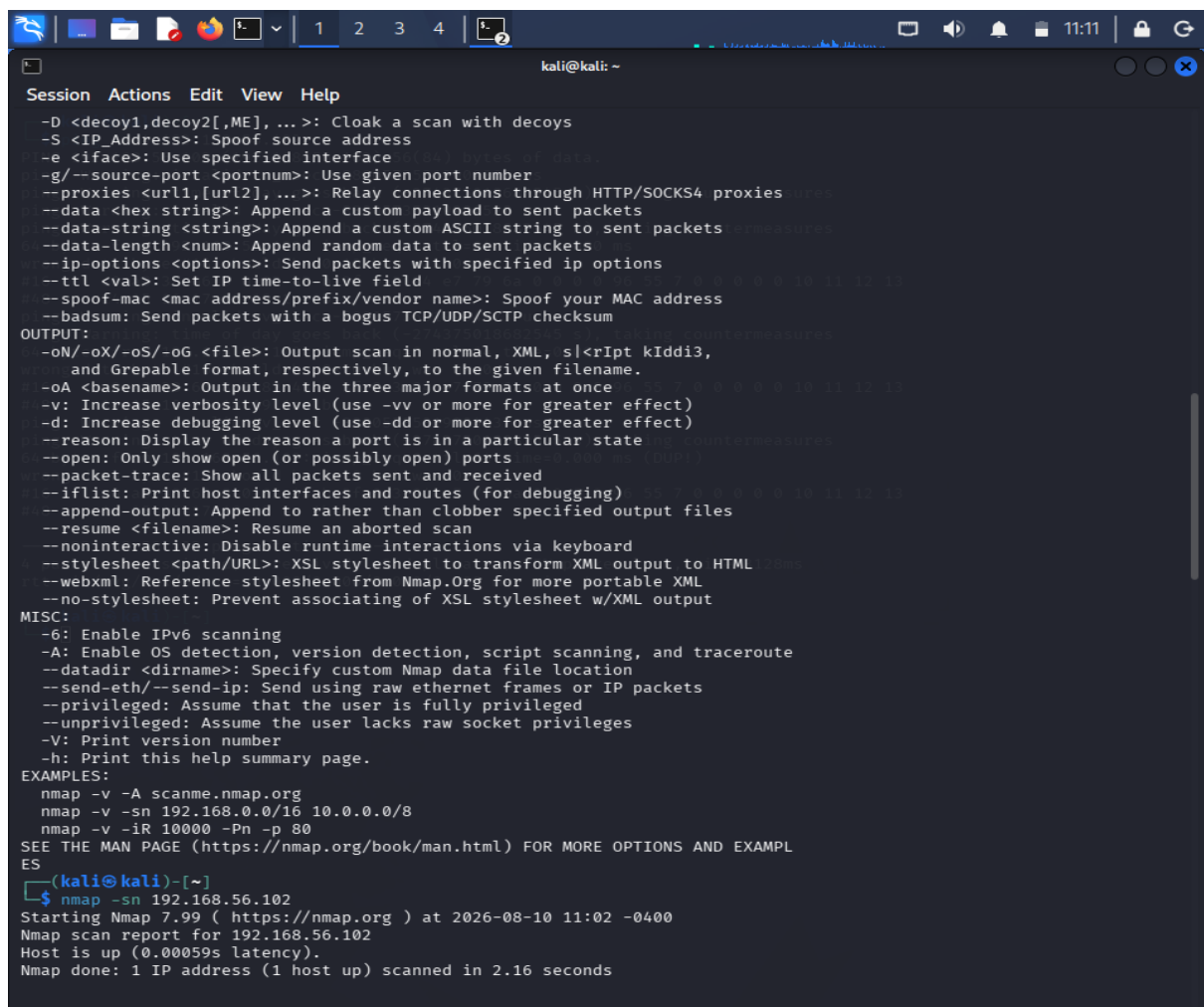
nmap -p- 192.168.56.102

nmap -sV 192.168.56.102

sudo nmap -O 192.168.56.102

nmap -sC 192.168.56.102

sudo nmap -sC -sV -O 192.168.56.102



```
kali@kali: ~  
Session Actions Edit View Help  
-D <decoy1,decoy2[,ME],...>: Cloak a scan with decoys  
-S <IP_Address>: Spoof source address  
-e <iface>: Use specified interface  
-g/--source-port <portnum>: Use given port number  
--proxies <url1,[url2],...>: Relay connections through HTTP/SOCKS4 proxies  
--data <hex string>: Append a custom payload to sent packets  
--data-string <string>: Append a custom ASCII string to sent packets  
--data-length <num>: Append random data to sent packets  
--ip-options <options>: Send packets with specified ip options  
--ttl <val>: Set IP time-to-live field  
--spoof-mac <mac address/prefix/vendor name>: Spoof your MAC address  
--badsum: Send packets with a bogus TCP/UDP/SCTP checksum  
OUTPUT:  
-oN/-oX/-oS/-oG <file>: Output scan in normal, XML, sI<riPt kIddi3, and Grepable format, respectively, to the given filename.  
-oA <basename>: Output in the three major formats at once  
-v: Increase verbosity level (use -vv or more for greater effect)  
-d: Increase debugging level (use -dd or more for greater effect)  
--reason: Display the reason a port is in a particular state  
--open: Only show open (or possibly open) ports  
--packet-trace: Show all packets sent and received  
--iflist: Print host interfaces and routes (for debugging)  
--append-output: Append to rather than clobber specified output files  
--resume <filename>: Resume an aborted scan  
--noninteractive: Disable runtime interactions via keyboard  
--stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML  
--webxml: Reference stylesheet from Nmap.Org for more portable XML  
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output  
MISC:  
-6: Enable IPv6 scanning  
-A: Enable OS detection, version detection, script scanning, and traceroute  
--datadir <dirname>: Specify custom Nmap data file location  
--send-eth/--send-ip: Send using raw ethernet frames or IP packets  
--privileged: Assume that the user is fully privileged  
--unprivileged: Assume the user lacks raw socket privileges  
-V: Print version number  
-h: Print this help summary page.  
EXAMPLES:  
nmap -v -A scanme.nmap.org  
nmap -v -sn 192.168.0.0/16 10.0.0.0/8  
nmap -v -iR 10000 -Pn -p 80  
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES  
(kali@kali)-[~]  
$ nmap -sn 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:02 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.00059s latency).  
Nmap done: 1 IP address (1 host up) scanned in 2.16 seconds
```

Host Discovery

The host discovery scan was performed to verify whether the Windows workstation was accessible from the assessment machine.

Target: 192.168.56.102

Host Status:

```
Session Actions Edit View Help
Nmap done: 1 IP address (1 host up) scanned in 2.16 seconds

(kali@kali)-[~]
$ nmap 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:02 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0011s latency).
All 1000 scanned ports on 192.168.56.102 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 5.82 seconds

(kali@kali)-[~]
$ nmap -p- 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:03 -0400
Nmap scan report for 192.168.56.102
Host is up (0.22s latency).
All 65535 scanned ports on 192.168.56.102 are in ignored states.
Not shown: 60791 filtered tcp ports (no-response), 4744 filtered tcp ports (net-unreach)
Nmap done: 1 IP address (1 host up) scanned in 222.14 seconds

(kali@kali)-[~]
$ nmap -sV 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:07 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0014s latency).
All 1000 scanned ports on 192.168.56.102 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.91 seconds

(kali@kali)-[~]
$ sudo nmap -O 192.168.56.102
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:08 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0019s latency).
All 1000 scanned ports on 192.168.56.102 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCsis cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.11 seconds
```

3.4 Port and Service Findings

Port	Service	Version	Purpose	Risk
[Actual]	[Actual]	[Actual]	[Purpose]	[Risk]
[Actual]	[Actual]	[Actual]	[Purpose]	[Risk]
[Actual]	[Actual]	[Actual]	[Purpose]	[Risk]

Windows Service Analysis

Particular attention was given to file-sharing and remote-management services.

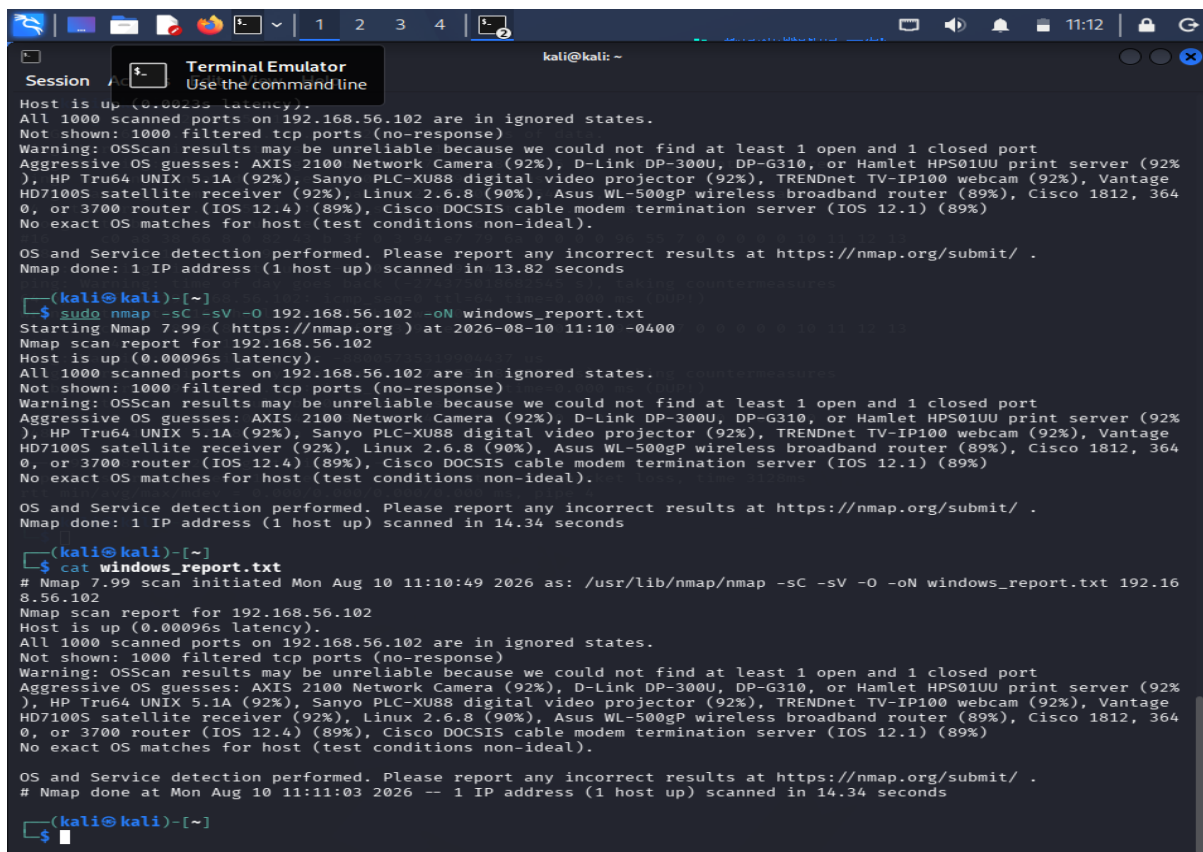
If identified by the scan, common examples include:

- **RPC:** Used by Windows for various remote procedure and management functions.
- **SMB:** Used for Windows file and printer sharing.
- **RDP:** Used for remote graphical administration.
- **WinRM:** Used for remote Windows management.

The actual report should list only the services identified by the student's scan.

Least Privilege Analysis

The principle of least privilege requires that only services necessary for the workstation's business function should be exposed.



```
kali@kali: ~  
Session  
Terminal Emulator  
Use the command line  
Host is up (0.0025s latency).  
All 1000 scanned ports on 192.168.56.102 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 13.82 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -sC -sV -O 192.168.56.102 -oN windows_report.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:10 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.00096s latency).  
All 1000 scanned ports on 192.168.56.102 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 14.34 seconds  
  
(kali@kali)-[~]  
$ cat windows_report.txt  
# Nmap 7.99 scan initiated Mon Aug 10 11:10:49 2026 as: /usr/lib/nmap/nmap -sC -sV -O -oN windows_report.txt 192.168.56.102  
Nmap scan report for 192.168.56.102  
Host is up (0.00096s latency).  
All 1000 scanned ports on 192.168.56.102 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
# Nmap done at Mon Aug 10 11:11:03 2026 -- 1 IP address (1 host up) scanned in 14.34 seconds  
  
(kali@kali)-[~]  
$
```

Unnecessary SMB, RDP, remote-management, or other network services can increase the attack surface. These services should be accessible only to authorized users and trusted systems.

Security Implications

An attacker who obtains information about Windows services can use it for further reconnaissance. Exposed file-sharing or remote-management services may provide opportunities for unauthorized access or lateral movement if they are incorrectly configured or inadequately protected.

Therefore, the workstation should not expose unnecessary services before being connected to a production network.

Hardening Recommendations

1. Disable unnecessary Windows services.
2. Close unused firewall ports.
3. Restrict SMB access to trusted networks.
4. Disable legacy and insecure protocols where they are not required.
5. Restrict RDP to authorized administrators.
6. Secure WinRM and limit it to authorized management systems.
7. Apply Windows security updates.

8. Use strong authentication and account controls.
9. Monitor remote login activity.
10. Apply network segmentation where appropriate.

Conclusion

The enumeration of the Windows workstation provides information about its network exposure and helps determine whether it follows the principle of least privilege. Unnecessary services should be removed or restricted before the workstation is connected to the production environment.

QUESTION 4

Enumeration of OWASP Juice Shop Server – 192.168.56.103

Objective

The objective is to perform detailed network enumeration of the OWASP Juice Shop server at 192.168.56.103 before penetration testing begins. The assessment focuses on host discovery, port scanning, service-version detection, OS fingerprinting, and web-service identification.

Commands Executed

```
ping -c 4 192.168.56.103
nmap -sn 192.168.56.103
nmap 192.168.56.103
nmap -p- 192.168.56.103
nmap -sV 192.168.56.103
sudo nmap -O 192.168.56.103
nmap -sC 192.168.56.103
sudo nmap -sC -sV -O 192.168.56.103
```

Host Discovery

Nmap host discovery was performed to verify that the Juice Shop server was reachable.

Target: 192.168.56.103

Host Status:

```
kali@kali: ~  
Session Actions Edit View Help  
-(kali@kali)-[~]  
$ ping -c 4 192.168.56.103  
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.  
ping: Warning: invalid tv_usec -88005735319521724 us  
ping: Warning: time of day goes back (-218050817562025 s), taking countermeas  
ures  
ping: Warning: invalid tv_usec -88005735319520147 us  
ping: Warning: time of day goes back (-218050817562025 s), taking countermeas  
ures  
64 bytes from 192.168.56.103: icmp_seq=0 ttl=64 time=0.000 ms  
wrong data byte #16 should be 0x10 but was 0xc0  
#16  c0 a8 38 67 8 0 42 73 b 40 0 3 9c ea 79 6a 0 0 0 0 cc 21 9 0 0 0 0 0  
10 11 12 13  
#48  14 15 16 17 18 19 1a 1b  
ping: Warning: invalid tv_usec -88005735319519916 us  
ping: Warning: time of day goes back (-218050817562025 s), taking countermeas  
ures  
64 bytes from 192.168.56.103: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)  
wrong data byte #16 should be 0x10 but was 0xc0  
#16  c0 a8 38 67 8 0 42 73 b 40 0 3 9c ea 79 6a 0 0 0 0 cc 21 9 0 0 0 0 0  
10 11 12 13  
#48  14 15 16 17 18 19 1a 1b  
ping: Warning: invalid tv_usec -88005735319519737 us  
ping: Warning: time of day goes back (-218050817562025 s), taking countermeas  
ures  
64 bytes from 192.168.56.103: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)  
wrong data byte #16 should be 0x10 but was 0xc0  
#16  c0 a8 38 67 8 0 42 73 b 40 0 3 9c ea 79 6a 0 0 0 0 cc 21 9 0 0 0 0 0  
10 11 12 13  
#48  14 15 16 17 18 19 1a 1b  
— 192.168.56.103 ping statistics —  
4 packets transmitted, 1 received, +2 duplicates, 75% packet loss, time 3173m  
s  
rtt min/avg/max/mdev = 0.000/0.000/0.000/0.000 ms, pipe 4  
-(kali@kali)-[~]  
$
```

```
kali@kali: ~  
Session Actions Edit View Help  
--ttl <val>: Set IP time-to-live field  
--spoof-mac <mac address/prefix/vendor name>: Spoof your MAC address  
--badsum: Send packets with a bogus TCP/UDP/SCTP checksum  
OUTPUT:  
-oN/-oX/-oS/-oG <file>: Output scan in normal, XML, sI<R>pt Kiddi3, nmapme  
and Grepable format, respectively, to the given filename.  
-oA <basename>: Output in the three major formats at once  
-v: Increase verbosity level (use -vv or more for greater effect)  
-d: Increase debugging level (use -dd or more for greater effect)  
--reason: Display the reason a port is in a particular state  
--open: Only show open (or possibly open) ports  
--packet-trace: Show all packets sent and received  
--iflist: Print host interfaces and routes (for debugging)  
--append-output: Append to rather than clobber specified output files  
--resume <filename>: Resume an aborted scan  
--noninteractive: Disable runtime interactions via keyboard  
--stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML  
--webxml: Reference stylesheet from Nmap.Org for more portable XML  
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output  
MISC:  
-6: Enable IPv6 scanning  
-A: Enable OS detection, version detection, script scanning, and traceroute  
--datadir <dirname>: Specify custom Nmap data file location  
--send-eth/--send-ip: Send using raw ethernet frames or IP packets  
--privileged: Assume that the user is fully privileged  
--unprivileged: Assume the user lacks raw socket privileges  
-V: Print version number  
-h: Print this help summary page.  
EXAMPLES:  
nmap -v -sn scanme.nmap.org  
nmap -v -sn 192.168.0.0/16 10.0.0.0/8  
nmap -v -iR 10000 -Pn -p 80  
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPL  
ES  
-(kali@kali)-[~]  
$ nmap -sn 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:14 -0400  
Nmap scan report for 192.168.56.103  
Host is up (0.0026s latency).  
Nmap done: 1 IP address (1 host up) scanned in 0.79 seconds  
-(kali@kali)-[~]  
$ nmap 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:15 -0400  
Nmap scan report for 192.168.56.103  
Host is up (0.0023s latency).  
All 1000 scanned ports on 192.168.56.103 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Nmap done: 1 IP address (1 host up) scanned in 4.76 seconds
```

```
Session / Terminal Emulator
Use the command line
kali@kali: ~
Host is up (0.0023s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 4.76 seconds
(kali@kali)-[~]
$ nmap -p- 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:15 -0400
(kali@kali)-[~]
$ nmap -sV 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:31 -0400
Nmap scan report for 192.168.56.103
Host is up (0.00098s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.98 seconds
(kali@kali)-[~]
$ sudo nmap -O 192.168.56.103
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:32 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0014s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS010U print server (92%), HP Tru64 UNIX 5.1A (92%), Senyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500GP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.02 seconds
(kali@kali)-[~]
$ nmap -sS 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:32 -0400
Nmap scan report for 192.168.56.103
Host is up (0.00097s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 14.35 seconds
```

Port and Service Findings

Port	Service	Version	Purpose	Risk
[Actual]	[Actual]	[Actual]	Web/application service	[Risk]
[Actual]	[Actual]	[Actual]	Supporting service	[Risk]
[Actual]	[Actual]	[Actual]	Management service	[Risk]

Web Service Identification

The web service was further examined using Nmap's HTTP scripts where applicable.

For example:

```
nmap -p 80 --script http-title,http-headers 192.168.56.103
```

or, if HTTPS is exposed:

```
nmap -p 443 --script http-title,http-headers 192.168.56.103
```

The results can provide information such as the HTTP title and response headers.

Analysis of Exposed Services

The web service is essential for delivering the Juice Shop application to users. However, unnecessary administrative interfaces, databases, or supporting services should not be exposed to untrusted networks.

The enumeration information can assist a penetration tester in understanding the application environment and prioritizing further authorized testing.


```
kali@kali: ~  
Session Actions Edit View Help  
Nmap done: 1 IP address (1 host up) scanned in 14.35 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -sC -sV -O 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:33 -0400  
Nmap scan report for 192.168.56.103  
Host is up (0.0019s latency).  
All 1000 scanned ports on 192.168.56.103 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 17.73 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -sC -sV -O 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:35 -0400  
Nmap scan report for 192.168.56.103  
Host is up (0.0018s latency).  
All 1000 scanned ports on 192.168.56.103 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 13.68 seconds  
  
(kali@kali)-[~]  
$ nmap -p 80 --script http-title,http-headers 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:43 -0400  
Nmap scan report for 192.168.56.103  
Host is up (0.00076s latency).  
  
PORT      STATE      SERVICE  
80/tcp    filtered  http  
  
Nmap done: 1 IP address (1 host up) scanned in 0.95 seconds  
  
(kali@kali)-[~]  
$ nmap -p 443 --script http-title,http-headers 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:43 -0400
```

```
kali@kali: ~  
Session Actions Edit View Help  
Nmap done: 1 IP address (1 host up) scanned in 0.95 seconds  
  
(kali@kali)-[~]  
$ nmap -p 443 --script http-title,http-headers 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:43 -0400  
Nmap scan report for 192.168.56.103  
Host is up (0.0015s latency).  
  
PORT      STATE SERVICE  
443/tcp   filtered https  
  
Nmap done: 1 IP address (1 host up) scanned in 1.39 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -sC -sV -O 192.168.56.103 -oN juiceshop_report.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:44 -0400  
Nmap scan report for 192.168.56.103  
Host is up (0.00053s latency).  
All 1000 scanned ports on 192.168.56.103 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 13.18 seconds  
  
(kali@kali)-[~]  
$ cat juiceshop_report.txt  
# Nmap 7.99 scan initiated Mon Aug 10 11:44:42 2026 as: /usr/lib/nmap/nmap -sC -sV -O -oN juiceshop_report.txt 192.168.56.103  
Nmap scan report for 192.168.56.103  
Host is up (0.00053s latency).  
All 1000 scanned ports on 192.168.56.103 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
# Nmap done at Mon Aug 10 11:44:55 2026 -- 1 IP address (1 host up) scanned in 13.18 seconds  
  
(kali@kali)-[~]  
$
```

Security Risks

Potential risks associated with exposed web infrastructure include:

- Unnecessary network services
- Outdated web-server software
- Weak configurations
- Exposed administrative interfaces
- Information disclosure through service banners
- Unrestricted supporting services
- Inadequate access controls

Recommendations

1. Expose only the ports required by the application.
2. Restrict administrative interfaces.
3. Keep web-server and application software patched.
4. Use HTTPS for production communication.
5. Restrict databases and internal supporting services.

6. Use firewall rules to reduce unnecessary exposure.
7. Remove unnecessary service information disclosure.
8. Monitor application and web-server logs.

Conclusion

The enumeration of the OWASP Juice Shop server provides an understanding of its network exposure before penetration testing. Identifying web services, supporting services, versions, and the operating system helps testers prioritize authorized security assessments while helping administrators minimize unnecessary exposure.

QUESTION 5

Enumeration of Network Gateway – 192.168.56.1

Objective

The objective is to enumerate the laboratory network gateway at 192.168.56.1 and identify exposed network and management services. Particular attention is given to SSH, Telnet, SNMP, HTTP, and HTTPS because management interfaces can significantly affect the security of network infrastructure.

Commands Executed

```
ping -c 4 192.168.56.1
nmap -sn 192.168.56.1
nmap 192.168.56.1
nmap -p- 192.168.56.1
nmap -sV 192.168.56.1
sudo nmap -O 192.168.56.1
nmap -sC 192.168.56.1
sudo nmap -sC -sV -O 192.168.56.1
For SNMP:
sudo nmap -sU -p 161 -sV 192.168.56.1
```

Host Discovery

Target: 192.168.56.1

Host Status:

```
kali@kali: ~  
$ ping -c 4 192.168.56.1  
PING 192.168.56.1 (192.168.56.1) 56(84) bytes of data:  
64 bytes from 192.168.56.1: icmp_seq=1 ttl=64 time=2.69 ms  
64 bytes from 192.168.56.1: icmp_seq=2 ttl=64 time=1.34 ms  
64 bytes from 192.168.56.1: icmp_seq=3 ttl=64 time=2.00 ms  
64 bytes from 192.168.56.1: icmp_seq=4 ttl=64 time=1.35 ms  
--- 192.168.56.1 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3584ms  
rtt min/avg/max/mdev = 1.343/1.845/2.694/0.558 ms  
$ nmap -sn 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 12:00 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00065s latency).  
Nmap done: 1 IP address (1 host up) scanned in 0.56 seconds  
$ nmap 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 12:00 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.0010s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
Nmap done: 1 IP address (1 host up) scanned in 5.45 seconds  
$ nmap -sV 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 12:00 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.0012s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
135/tcp   open  msrpc        Microsoft Windows RPC  
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds?   
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 11.67 seconds  
$ sudo nmap -O 192.168.56.1
```

```
kali@kali: ~  
Session Actions Edit View Help  
[kali@kali]~  
$ nmap -sV 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 12:00 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.0012s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
135/tcp   open  msrpc        Microsoft Windows RPC  
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds?  
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 11.67 seconds  
  
[kali@kali]~  
$ sudo nmap -O 192.168.56.1  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 12:01 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.0015s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
135/tcp   open  msrpc        Microsoft Windows RPC  
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds?  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode ne  
twork gateway (90%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 10.48 seconds  
  
[kali@kali]~  
$ nmap -sC 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 12:03 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00065s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
135/tcp   open  msrpc        Microsoft Windows RPC  
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds?  
  
Host script results:  
| smb2-time:  
|_ 3.1.1:  
|_ Message signing enabled and required  
|_ smb2-time:  
|_ date: 2026-08-10T16:09:32  
|_ start_date: N/A  
|_ nbstat: NetBIOS name: DESKTOP-5RH1053, NetBIOS user: <unknown>, NetBIOS MAC: 0a:00:27:00:00:0f (unknown)
```

```
kali@kali: ~  
Session Actions Edit View Help  
Terminal Emulator  
Use the command line  
Nmap done: 1 IP address (1 host up) scanned in 20.87 seconds  
  
[kali@kali]~  
$ sudo nmap -sC -sV -O 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 12:09 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.0012s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
135/tcp   open  msrpc        Microsoft Windows RPC  
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds?  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode ne  
twork gateway (90%)  
No exact OS matches for host (test conditions non-ideal).  
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows  
  
Host script results:  
| smb2-security-mode:  
|_ 3.1.1:  
|_ Message signing enabled and required  
|_ smb2-time:  
|_ date: 2026-08-10T16:09:32  
|_ start_date: N/A  
|_ nbstat: NetBIOS name: DESKTOP-5RH1053, NetBIOS user: <unknown>, NetBIOS MAC: 0a:00:27:00:00:0f (unknown)  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 22.20 seconds  
  
[kali@kali]~  
$ sudo nmap -sU -p 161 -sV 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 12:10 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00045s latency).  
  
PORT      STATE SERVICE VERSION  
161/udp   filtered snmp  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 0.79 seconds  
  
[kali@kali]~  
$ nmap -p 80 --script http-title,http-headers 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 12:11 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00076s latency).
```

Network Services Identified

Port	Protocol	Service	Version	Purpose	Risk
[Actual]	TCP/UDP	[Actual]	[Actual]	[Purpose]	[Risk]
[Actual]	TCP/UDP	[Actual]	[Actual]	[Purpose]	[Risk]
[Actual]	TCP/UDP	[Actual]	[Actual]	[Purpose]	[Risk]

Management Service Analysis

SSH – Port 22

SSH is commonly used for secure remote administration. Although it provides encrypted communication, it should not be exposed unnecessarily. Access should be restricted to authorized administrators and trusted management systems.

Telnet – Port 23

Telnet is an insecure remote-management protocol because it does not provide the same protection expected from modern encrypted administration. If Telnet is discovered, it should normally be disabled and replaced with SSH.

SNMP – UDP 161

SNMP is used for network monitoring and management. Poorly secured SNMP configurations can disclose information about network infrastructure. Access should be restricted, and stronger SNMP versions should be preferred where supported.

HTTP – Port 80

HTTP may provide a web-based management interface. If administrative functionality is exposed unnecessarily, it increases the attack surface. Management access should be restricted and HTTPS should be preferred.

HTTPS – Port 443

HTTPS provides encrypted web management. However, the management interface should still be restricted to authorized administrators, and the gateway software should be kept patched.

Security Implications

An attacker who discovers exposed management interfaces can use this information for further reconnaissance and may attempt to exploit weak configurations, outdated services, poor authentication, or unnecessary network exposure.

A gateway is particularly important because compromise of network infrastructure can affect multiple systems behind it.

Recommendations

1. Disable Telnet where it is not required.

2. Use SSH for secure command-line administration.
3. Restrict SSH to trusted management networks.
4. Restrict SNMP to authorized monitoring systems.
5. Prefer secure SNMP configurations such as SNMPv3 where supported.
6. Disable HTTP management if unnecessary.
7. Use secure HTTPS configurations for web management.
8. Apply gateway firmware and security updates.
9. Use ACLs/firewall rules to limit management access.
10. Monitor administrator logins and configuration changes.
11. Use strong administrator authentication.
12. Regularly review exposed services.

Conclusion

The enumeration of the network gateway provides information about its exposed management interfaces and network services. Unnecessary management services can increase the risk of unauthorized access and compromise of network infrastructure. Restricting management access, disabling insecure protocols, maintaining updated firmware, and applying strong authentication are essential controls for reducing the gateway's attack surface.